

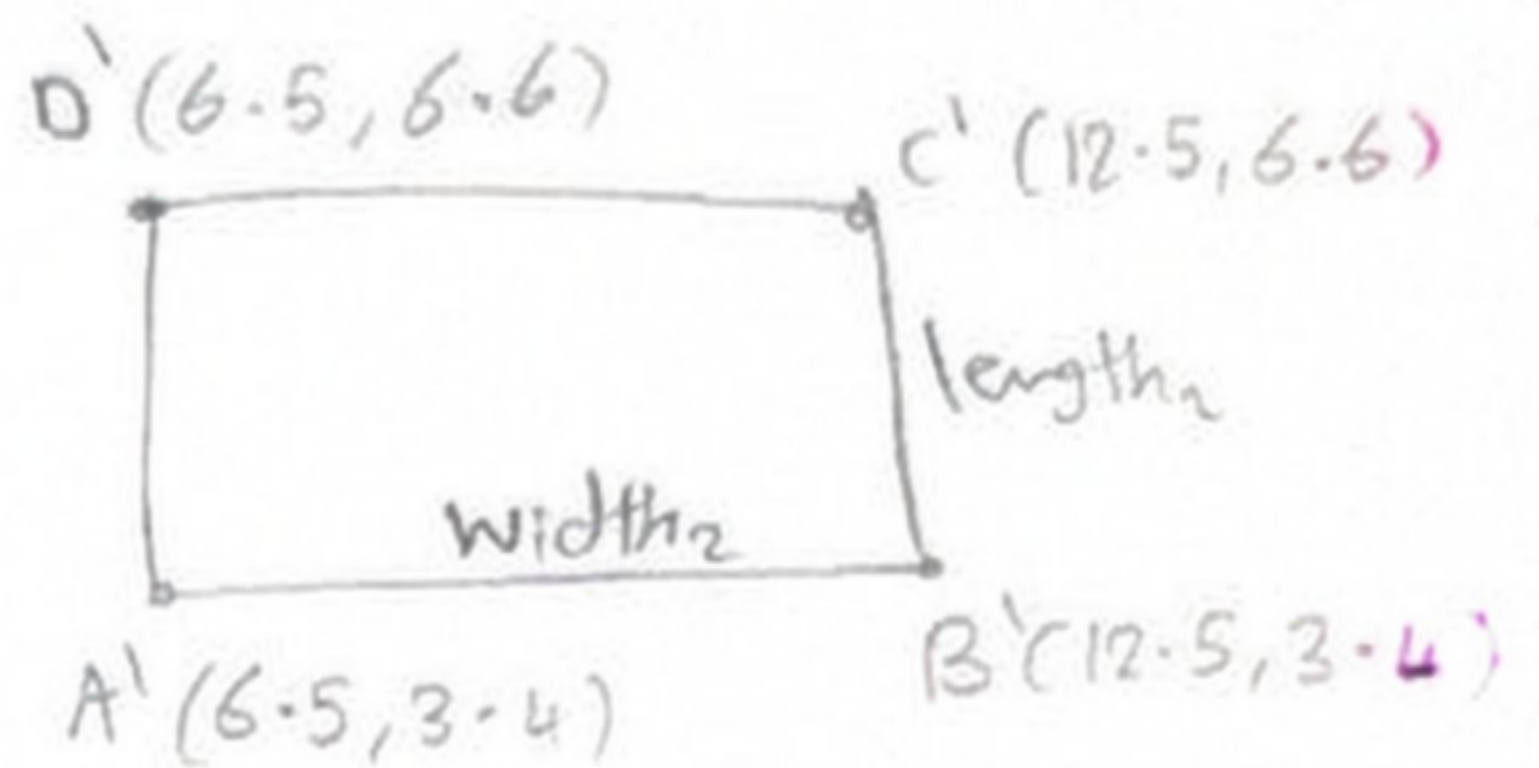
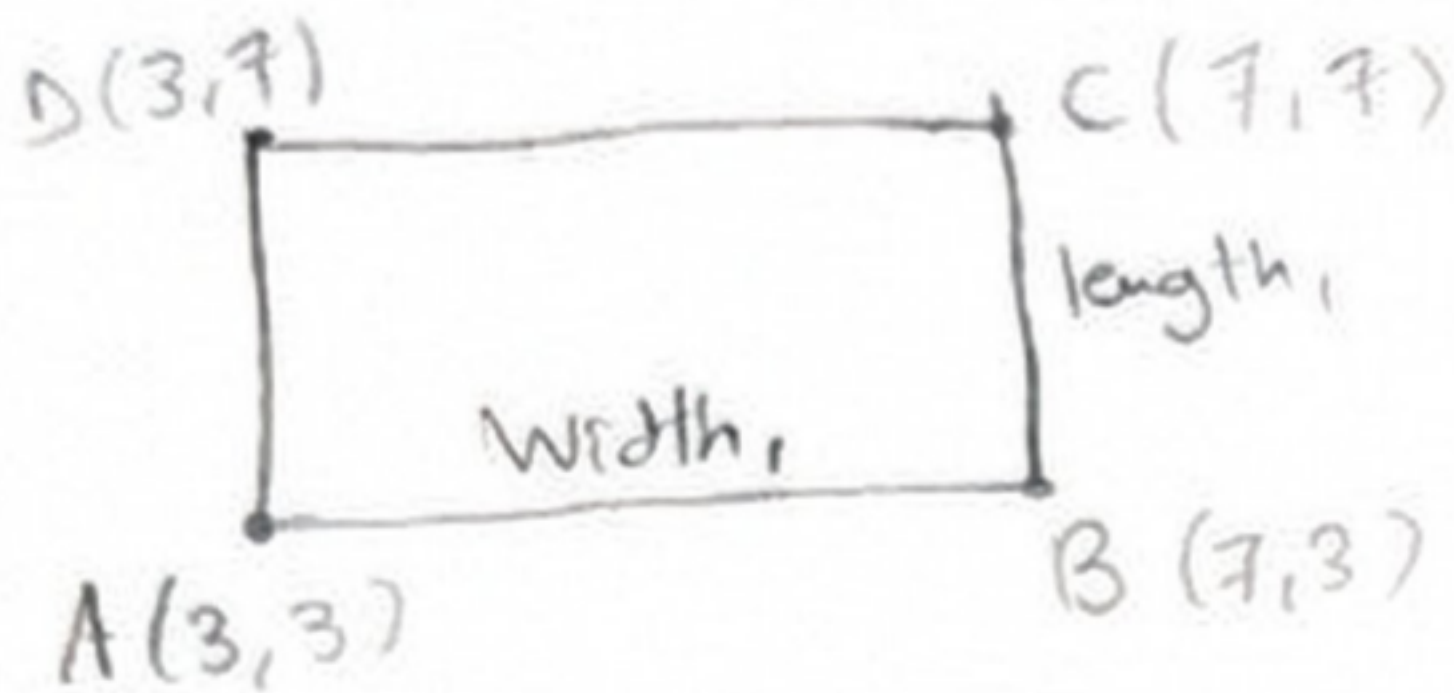
Ziad Ahmed Nader
22P0059

Midterm

~~17.5~~
~~20~~

28-3-2026

Question (2):



$$\text{width}_1 = 4$$
$$\text{width}_2 = 6$$

$$\text{length}_1 = 4$$
$$\text{length}_2 = 3.2$$

$$6 = 4 S_x$$

$$3.2 = 4 S_y$$

$$S_x = \frac{3}{2}$$

$$S_y = \frac{4}{5}$$

scaling factors

$$x' = S_x x + t_x$$

$$6.5 = \frac{3}{2}(3) + t_x$$

$$t_x = 2$$

$$y' = S_y y + t_y$$

$$6.6 = \frac{4}{5}(7) + t_y$$

$$t_y = 1$$

we perform scaling $\rightarrow$ translation

$$\therefore M = TS$$

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = \begin{bmatrix} \frac{4}{5} & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = TS = \begin{bmatrix} \frac{4}{5} & 0 & 1 \\ 0 & \frac{3}{2} & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix written wrong!

Question (3):

GE-TB-H3 → negative transformation is applied which switches the colors (white becomes black and black becomes white)

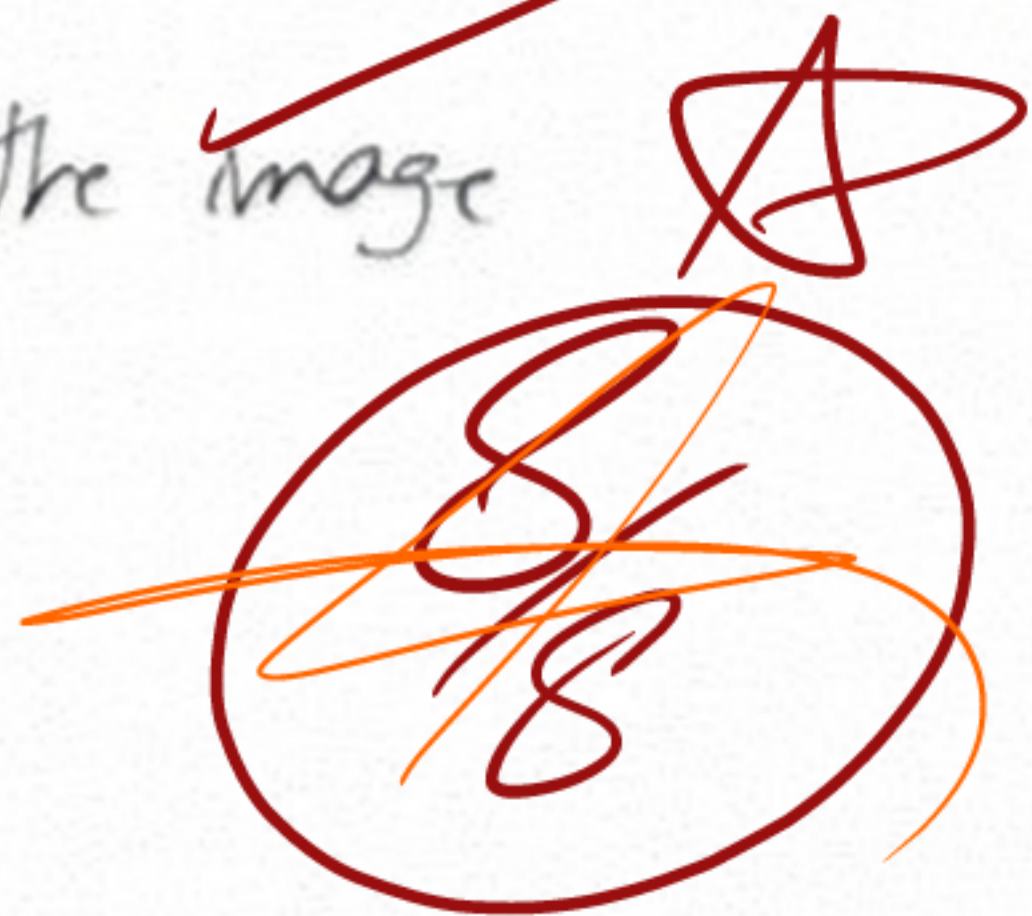
GF-TE-H5 → thresholding function that makes the image binary and have only two intensities (0, 255) in the Histogram

GC-TD-H1 → power law (Gamma $\gamma < 1$) which increases the overall intensities of the image and shift the Histogram right compared to original image

GB-TC-H4 → Power law (Gamma $\gamma > 1$) which darkens the overall image and increases the no. pixels that have low intensities in the Histogram

GD-TA-H2 → it increases a little bit the brightness of image which makes a small shift to the right in the high intensity values in Histogram

→ the unused graph is **A** because it has a slope 1 which doesn't make any change in the image



Question (4):

r_k	n_k	$P_r(r_k)$	S_k	S_k	$P_s(s_k)$
0	1	0.03125	0.21875 ~	0	0.03125
1	4	0.125	1.09375 ~	1	0.125
2	8	0.25	2.84375 ~	3	0.25
3	5	0.15625	3.9375 ~	4	} → 0.21875
4	2	0.0625	4.375 ~	4	
5	2	0.0625	4.8125 ~	5	0.0625
6	5	0.15625	5.90625 ~	6	0.15625
7	5	0.15625	7 ~	7	0.15625

z_q	n_q	$P_z(z_q)$	$G_z(z_q)$	$G_z(z_q)$
0	0	0	0 ~ 0	0
1	0	0	0 ~ 0	0
2	4	0.125	0.875 ~ 1	1
3	4	0.125	1.75 ~ 2	2
4	4	0.125	2.625 ~ 3	3
5	4	0.125	3.5 ~ 4	4
6	5	0.15625	4.59375 ~ 5	5
7	11	0.34375	7 ~ 7	7

mapped to
smallest
by convention

z_q	$G_z(z_q)$	S_k	$P_s(s_k)$
0 → 0	0	0	0.03125
1 → 0	0	1	0.125
2 → 1	1	3	0.25
3 → 2	2	4	} → 0.21875
4 → 3	3	4	
5 → 4	4	5	0.0625
6 → 5	5	6	0.15625
7 → 7	7	7	0.15625

z_q	Probability
0	0.03125
1	0.03125
2	0.125
3	0.125
4	0.25
5	0.21875
6	0.0625
7	0.015625

Q4): Histogram doesn't give spatial info of pixels
 image) 1.5 to -0.5

mapping isn't clear -1

$$\frac{3.5}{5}$$

Question (1):

Since we want a binarized image which means only two intensities, so we will make it black and white only to make image and text as clear as possible

1) Classification of intensity values of given Histogram

Intensity Values	no. pixels
0	10
50	25
100	20
150	55
200	25
250	300

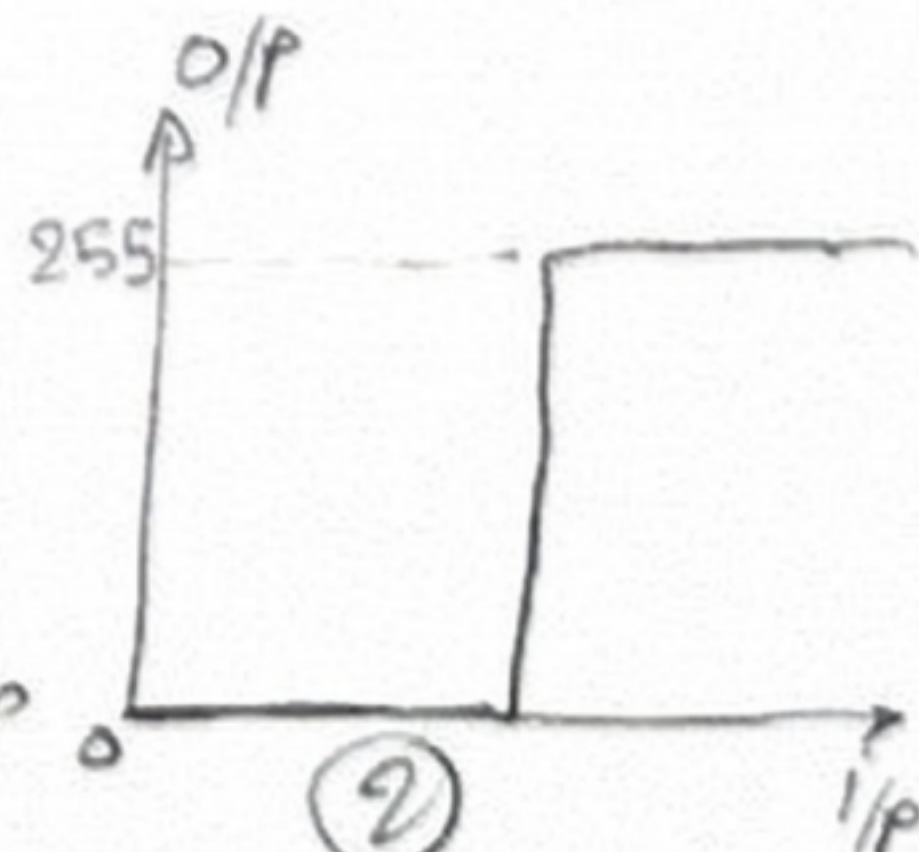
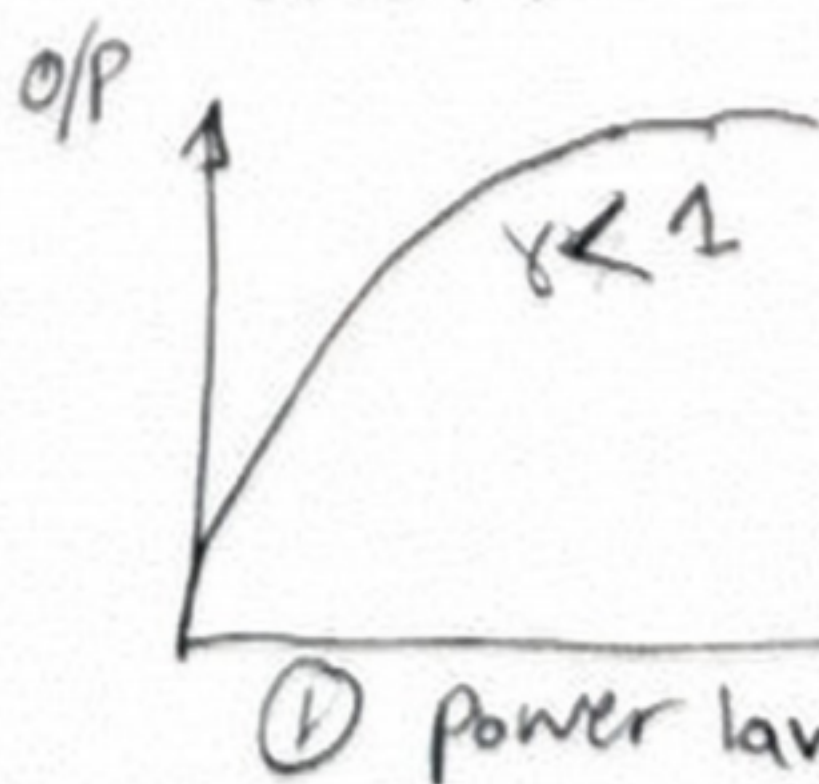
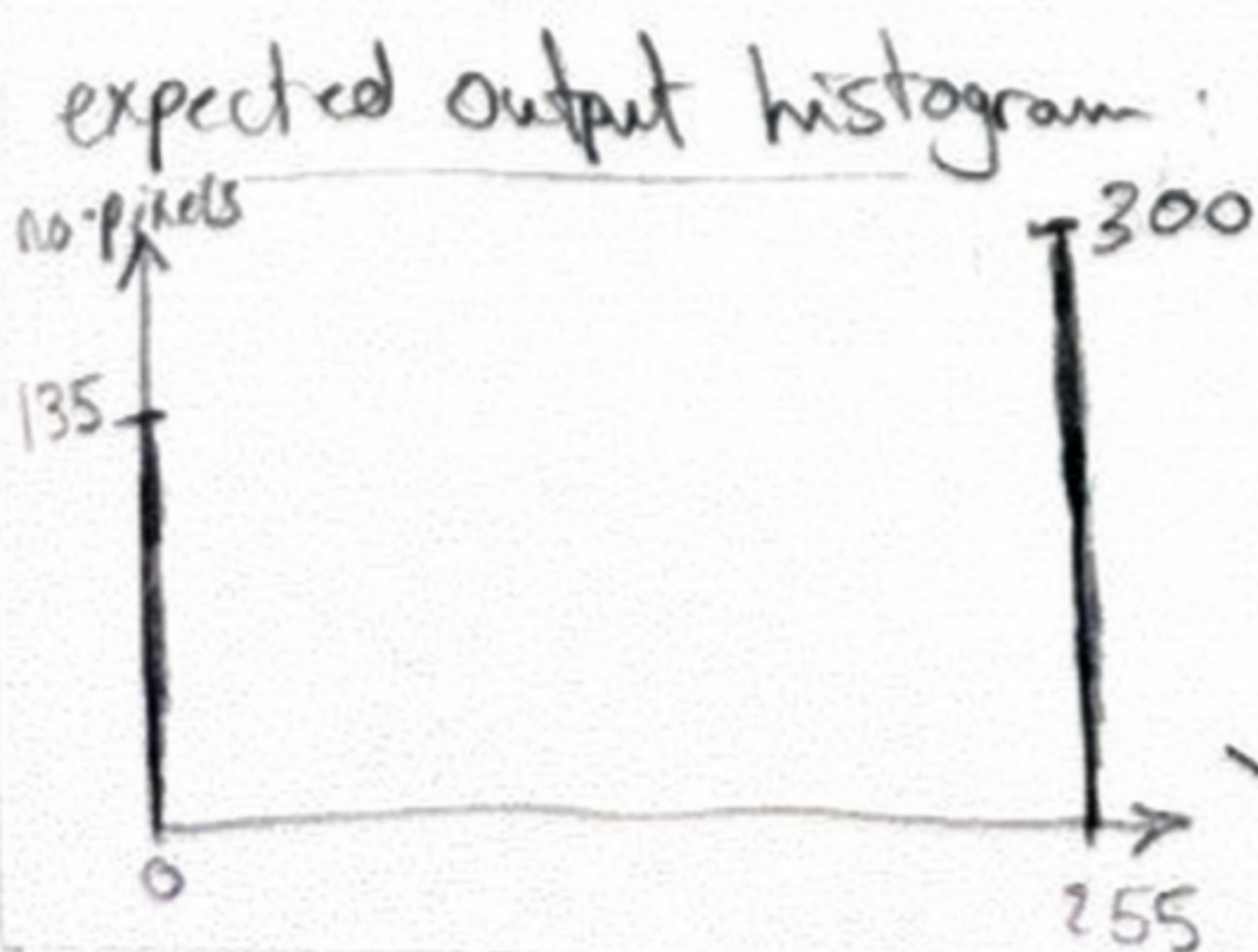
good but not practical also affects text

2) Techniques for enhancement

- power law ($\gamma < 1$) to increase the brightness of whole image (to remove shading)

- thresholding transform to keep only text black and background white

$$\frac{3.5}{4}$$



no. bright pixels > no. dark pixels because the white background